MATH1025B
Essential Mathematical Methods

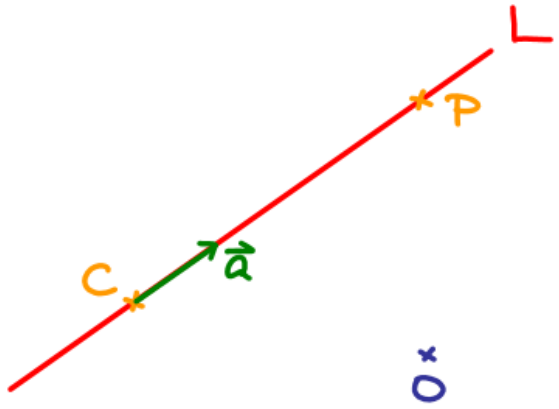
Chapter 10: Straight Lines, Planes and Curves

November 18, 2025

Straight Line in $\mathbb{R}^3$

Let

- $C = (c_1, c_2, c_3)$ be a fixed point
- $\vec{a} = (a_1, a_2, a_3)$ be a fixed vector (**direction vector**)
- L be a straight line passing through C and goes along direction $\vec{a}$
- $P = (x, y, z)$ be a movable point



Then, we have $\overrightarrow{CP} // \vec{a}$, i.e. $\overrightarrow{CP} = t\vec{a}$, or

$$(x - c_1, y - c_2, z - c_3) = t(a_1, a_2, a_3), \qquad t \in \mathbb{R}.$$

So we have the **parametric equation** of L:

$$\begin{cases} x = c_1 + ta_1 \\ y = c_2 + ta_2 \\ z = c_3 + ta_3 \end{cases}$$

Straight Line in $\mathbb{R}^3$

Parametric equation of L:

$$\begin{cases} x = c_1 + ta_1 \\ y = c_2 + ta_2 \\ z = c_3 + ta_3 \end{cases}$$

Eliminating t, we get the **symmetric equation** of L:

$$\frac{x - c_1}{a_1} = \frac{y - c_2}{a_2} = \frac{z - c_3}{a_3} \qquad (\text{if } a_1, a_2, a_3 \neq 0)$$

If, say $a_1, a_2 \neq 0$ but $a_3 = 0$, then the equation becomes:

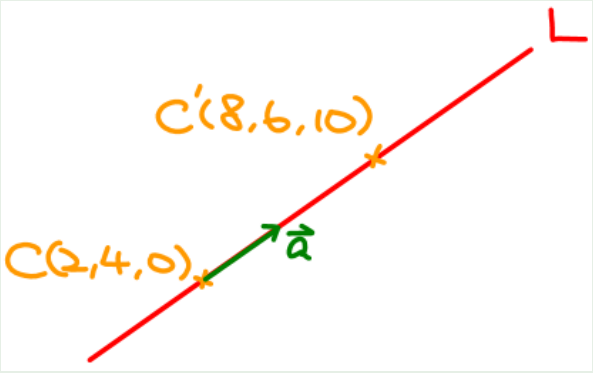
$$\frac{x - c_1}{a_1} = \frac{y - c_2}{a_2} \quad \text{and} \quad z = c_3.$$

Example 10.1

Suppose the equation of a straight line L in $\mathbb{R}^3$ is

$$\frac{x-2}{3} = y - 4 = \frac{z}{5}.$$

Then L passes through $(2, 4, 0)$ and goes along the direction $\vec{a} = (3, 1, 5)$.



However, L also passes through the point

$$C' = (2, 4, 0) + 2(3, 1, 5) = (8, 6, 10).$$

Therefore,

$$\frac{x-8}{3} = y - 6 = \frac{z-10}{5}$$

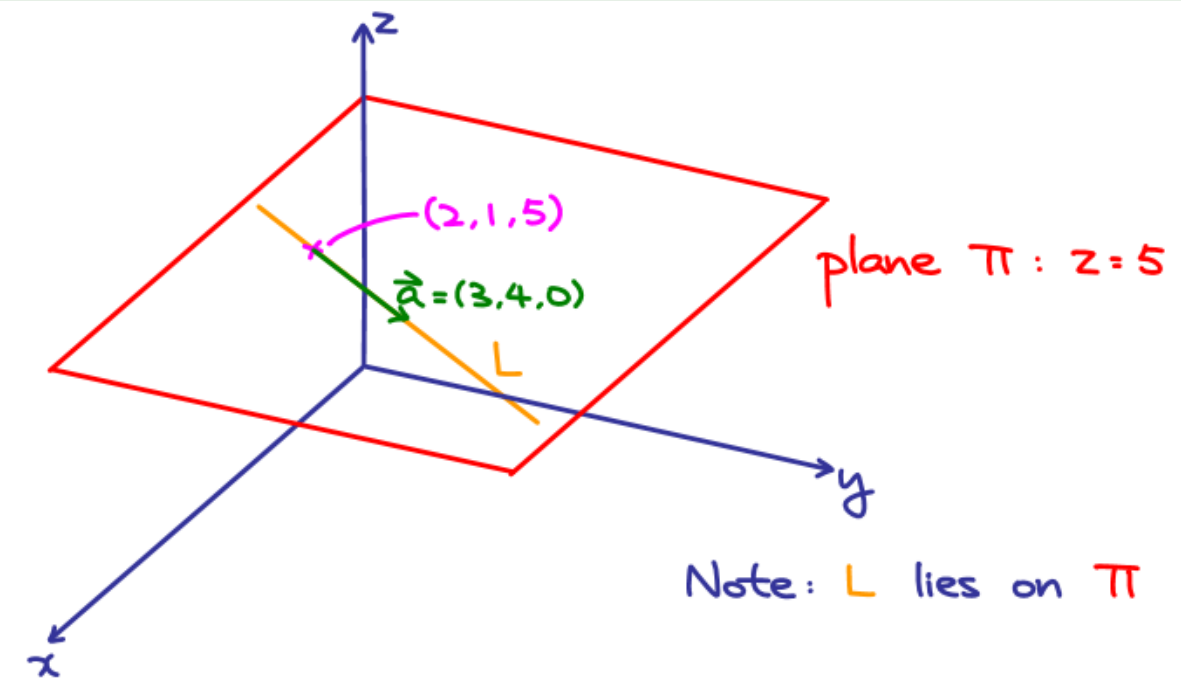
is also an equation of L.

Example 10.2

If the equation of a straight line L in $\mathbb{R}^3$ is

$$\frac{x-2}{3} = \frac{y-1}{4} \quad \text{and} \quad z = 5,$$

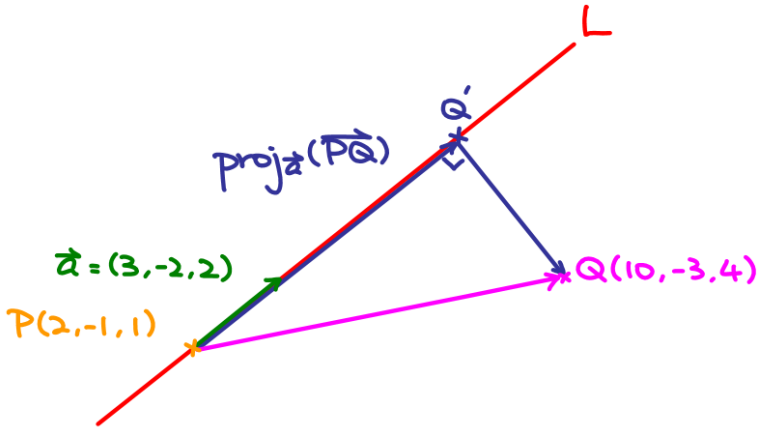
then L passes through $(2, 1, 5)$ and goes along the direction $\vec{a} = (3, 4, 0)$.



Example 10.3

Suppose L is a straight line in $\mathbb{R}^3$ given by the equation $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-2}{2}$, and $Q = (10, -3, 4)$ is a fixed point. What is the shortest distance between L and Q?

The shortest distance between L and Q is the distance between Q and Q', where Q' is the foot of perpendicular from Q to L.



- L passes through $P(2, -1, 1)$.
- Direction vector of L: $\vec{a} = (3, -2, 2)$
- $\overrightarrow{PQ} = (10, -3, 4) - (3, -2, 2) = (8, -2, 3)$
- $\overrightarrow{PQ'} = \text{proj}_{\vec{a}}(\overrightarrow{PQ}) = \frac{\overrightarrow{PQ} \cdot \vec{a}}{|\vec{a}|^2}\vec{a} = \frac{34}{17}\vec{a} = 2\vec{a} = (6, -4, 4)$
- $\overrightarrow{Q'Q} = \overrightarrow{PQ} - \overrightarrow{PQ'} = (8, -2, 3) - (6, -4, 4) = (2, 2, -1)$

The shortest distance between L and Q: $|\overrightarrow{QQ'}| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$.

Straight Lines in $\mathbb{R}^3$

In general, if L is a straight line in $\mathbb{R}^n$ which

- passes through a fixed point $\vec{c} = (c_1, c_2, \cdots, c_n)$ and
- goes along the direction $\vec{a} = (a_1, a_2, \ldots, a_n)$,

then a parametric equation of L is

$$\vec{x} = \vec{c} + t\vec{a}, \quad t \in \mathbb{R},$$

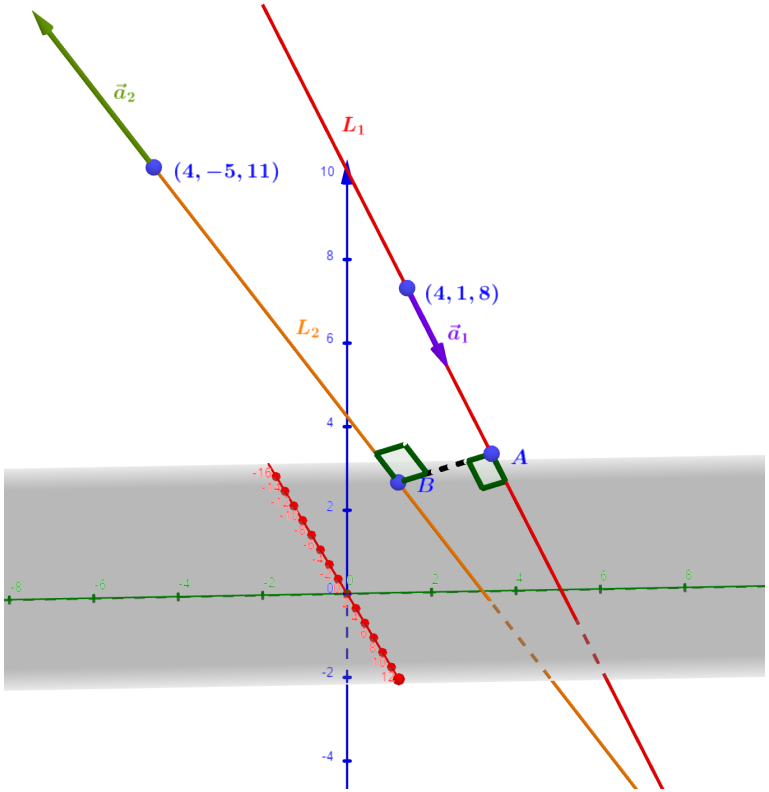
where $\vec{x} = (x_1, x_2, \ldots, x_n)$.
If $a_i \neq 0$ for all i, by eliminating t, we obtain

$$\frac{x_1 - c_1}{a_1} = \frac{x_2 - c_2}{a_2} = \cdots = \frac{x_n - c_n}{a_n}.$$

(What does the equation look like if some $a_i = 0$?)

Example 10.4

Let $\vec{c}_1 = (4, 1, 8), \vec{a}_1 = (0, 1, -2)$, $\vec{c}_2 = (4, -5, 11), \vec{a}_2 = (1, -3, 4)$.
Let $L_1 : \vec{x} = \vec{c}_1 + t\vec{a}_1$ and $L_2 : \vec{x} = \vec{c}_2 + s\vec{a}_2$, $t, s \in \mathbb{R}$, be two straight lines in $\mathbb{R}^3$. Find the shortest distance between L_1 and L_2.



Need to find A on L_1 and B on L_2 such that $\overrightarrow{BA}$ is perpendicular to both L_1 and L_2. Then the shortest distance between L_1 and L_2 is $|\overrightarrow{BA}|$.

Example 10.4

Let $\vec{c}_1 = (4, 1, 8), \vec{a}_1 = (0, 1, -2)$, $\vec{c}_2 = (4, -5, 11), \vec{a}_2 = (1, -3, 4)$.
Let $L_1 : \vec{x} = \vec{c}_1 + t\vec{a}_1$ and $L_2 : \vec{x} = \vec{c}_2 + s\vec{a}_2$, $t, s \in \mathbb{R}$, be two straight lines in $\mathbb{R}^3$. Find the shortest distance between L_1 and L_2.

Let $\overrightarrow{OA} = \vec{c}_1 + t_0\vec{a}_1$, $\overrightarrow{OB} = \vec{c}_2 + s_0\vec{a}_2$ for some $t_0, s_0 \in \mathbb{R}$.
Then

$$\begin{aligned}\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} &= (\vec{c}_1 - \vec{c}_2) + t_0\vec{a}_1 - s_0\vec{a}_2 \\ &= (0, 6, -3) + t_0(0, 1, -2) - s_0(1, -3, 4) \\ &= (-s_0, 6 + t_0 + 3s_0. - 3 - 2t_0 - 4s_0).\end{aligned}$$

Note $\overrightarrow{BA} \perp \vec{a}_1$ and $\overrightarrow{BA} \perp \vec{a}_2$ give two equations:

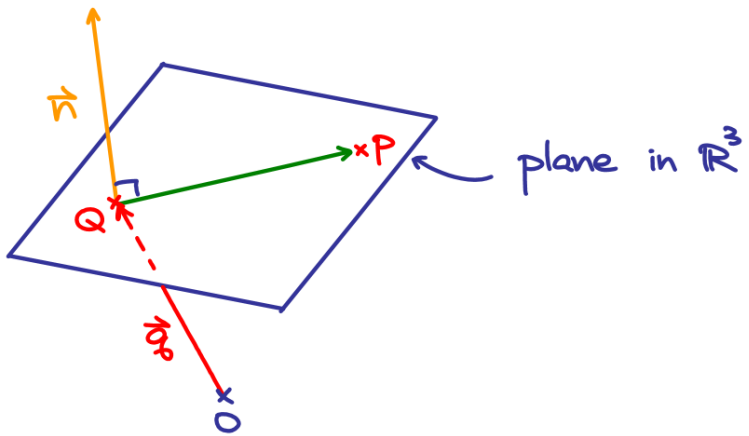
$$\begin{aligned}\overrightarrow{BA} \cdot \vec{a}_1 = 0 &\iff 12 + 5t_0 + 11s_0 = 0 \\ \overrightarrow{BA} \cdot \vec{a}_2 = 0 &\iff -30 - 11t_0 - 26s_0 = 0.\end{aligned}$$

Solving the equations, we get $t_0 = 2$ and $s_0 = -2$, and thus $\overrightarrow{BA} = (2, 2, 1)$.
The shortest distance between L_1 and L_2 is $|\overrightarrow{BA}| = \sqrt{2^2 + 2^2 + 1^2} = 3$.

Planes in $\mathbb{R}^3$

Given a plane in $\mathbb{R}^3$, let

- $Q = (q_1, q_2, q_3)$ be a fixed point on the plane
- $\vec{n} = (A, B, C)$ be a normal of the plane
- $P = (x, y, z)$ be a movable point on the plane



Then, we have $\vec{n} \perp \overrightarrow{QP}$, i.e. $\vec{n} \cdot \overrightarrow{QP} = 0$.

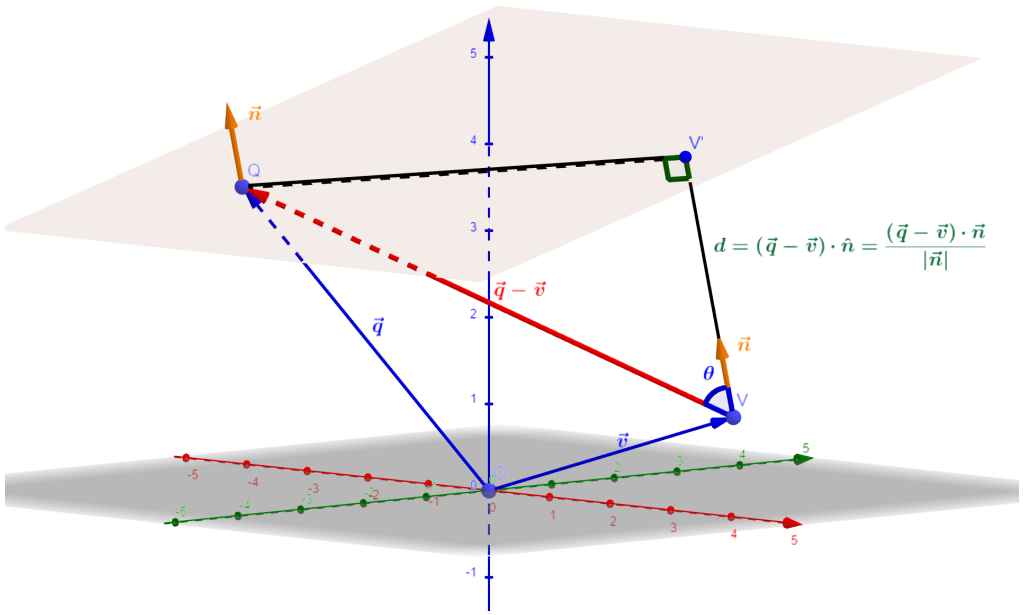
$$A(x - q_1) + B(y - q_2) + C(z - q_3) = 0$$
$$Ax + By + Cz + \underbrace{(-Aq_1 - Bq_2 - Cq_3)}_{\text{denote it by } D} = 0$$

So the equation of a plane in $\mathbb{R}^3$ is of the form

$$Ax + By + Cz + D = 0$$

where $\vec{n} = A\vec{i} + B\vec{j} + C\vec{k}$ is a normal.

Furthermore, if d is the distance between a point $V(x_0, y_0, z_0)$ and the plane $\pi : Ax + By + Cz + D = 0$, then $d = \left| |\overrightarrow{VQ}| \cos\theta \right|$ where θ is the angle between $\vec{n}$ and $\overrightarrow{VQ}$.



$$d = \left| |\overrightarrow{VQ}| \cos\theta \right| = ||\vec{q} - \vec{v}| \cos\theta| = \left| \frac{|\vec{n}||\vec{q} - \vec{v}| \cos\theta}{|\vec{n}|} \right| = \frac{|\vec{n} \cdot (\vec{q} - \vec{v})|}{|\vec{n}|}$$

$$= \frac{|(A, B, C) \cdot (q_1 - x_0, q_2 - y_0, q_3 - z_0)|}{|\vec{n}|}$$

$$= \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}} \qquad \text{(Recall: } D = -Aq_1 - Bq_2 - Cq_3\text{)}$$

Example 10.5

$\pi : 2x - 2y - z - 3 = 0$ is a plane in $\mathbb{R}^3$ with a normal $2\vec{i} - 2\vec{j} - \vec{k}$.
The distance between O and π is

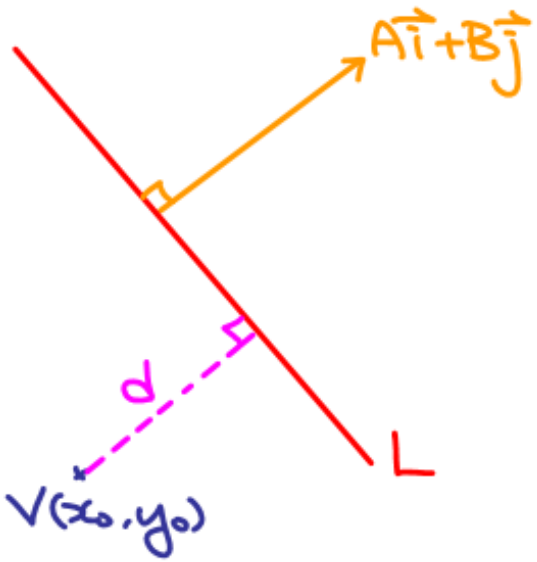
$$\frac{|2(0) - 2(0) - (0) - 3|}{\sqrt{2^2 + (-2)^2 + (-1)^2}} = 1.$$

Exercise 10.1 (Revisit of straight lines in $\mathbb{R}^2$)

Follow the idea of the discussion of planes in $\mathbb{R}^3$, show that if $L : Ax + By + C = 0$ is a straight line in $\mathbb{R}^2$, then

(a) $\vec{n} = A\vec{i} + B\vec{j}$ gives a normal of L;

(b) the distance between $V(x_0, y_0)$ and L is $d = \frac{|Ax_0+By_0+C|}{\sqrt{A^2+B^2}}$.



Example 10.6

Let $L : \dfrac{x-1}{2} = \dfrac{y-2}{-1} = \dfrac{z}{2}$ be a straight line and $\pi : x + y + z = 0$ be a plane in $\mathbb{R}^3$.

(a) Find the intersection of L and π.

(b) Find the angle between L and π.

(c) Find the projection of L on π.

(a) If P is a point lying on L, then

$$P = (1, 2, 0) + t(2, -1, 2) = (1 + 2t, 2 - t, 2t),\ t \in \mathbb{R}.$$

Suppose further that P lies on π, we have

$$\begin{aligned} (1 + 2t) + (2 - t) + (2t) &= 0 \\ 3t + 3 &= 0 \\ t &= -1. \end{aligned}$$

So L and π intersect at $(-1, 3, -2)$.

Example 10.6

Let $L : \dfrac{x-1}{2} = \dfrac{y-2}{-1} = \dfrac{z}{2}$ be a straight line and $\pi : x + y + z = 0$ be a plane in $\mathbb{R}^3$.

(b) Find the angle between L and π.

(b) Note $\vec{a} = (2, -1, 2)$ is a direction vector of L,
and $\vec{n} = (1, 1, 1)$ is a normal of π.
The angle between L and $\vec{n}$ is

$$\cos^{-1}\left(\frac{\vec{a}\cdot\vec{n}}{|\vec{a}||\vec{n}|}\right) = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right).$$

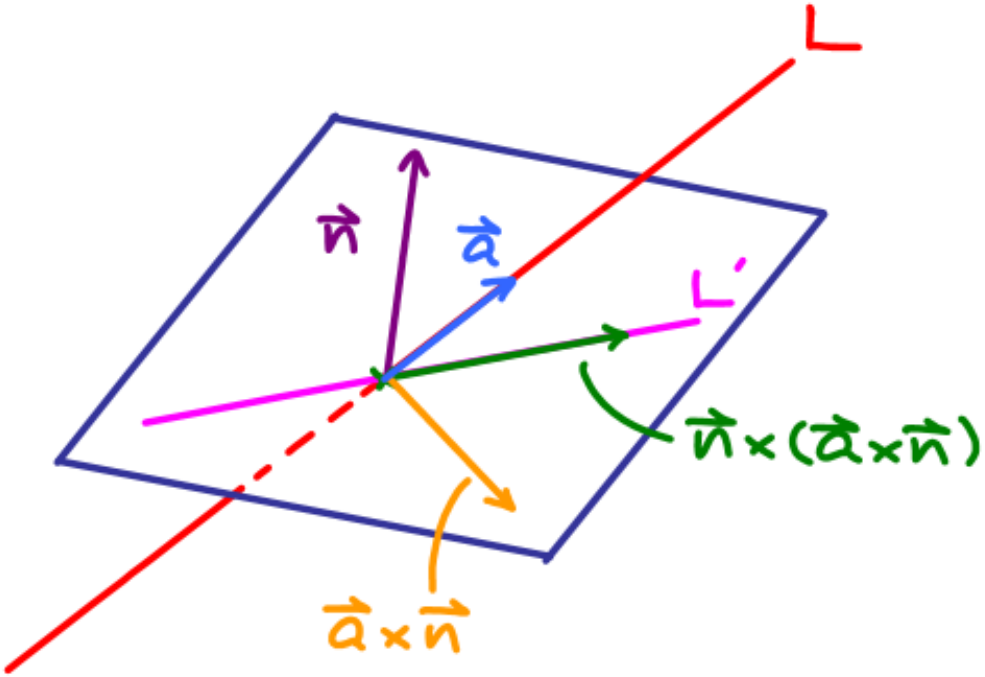
So the angle between L and π is $\dfrac{\pi}{2} - \cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$.

Example 10.6

Let $L : \dfrac{x-1}{2} = \dfrac{y-2}{-1} = \dfrac{z}{2}$ be a straight line and $\pi : x + y + z = 0$ be a plane in $\mathbb{R}^3$.

(c) Find the projection of L on π.

(c) Question: How to find a direction vector of the projection L'?



Example 10.6

Let $L : \dfrac{x-1}{2} = \dfrac{y-2}{-1} = \dfrac{z}{2}$ be a straight line and $\pi : x + y + z = 0$ be a plane in $\mathbb{R}^3$.

(c) Find the projection of L on π.

(c) Suppose L' has a direction vector $\vec{b}$.
Then $\vec{b}$ lies on the plane spanned by $\vec{n}$ and $\vec{a}$. So $\vec{b} \perp \vec{a} \times \vec{n}$.
Since $\vec{b}$ also lies on the plane π, we have $\vec{b} \perp \vec{n}$.
Hence $\vec{b} // \left(\vec{n} \times (\vec{a} \times \vec{n})\right)$.

$$\vec{n} \times (\vec{a} \times \vec{n}) = (\vec{i} + \vec{j} + \vec{k}) \times (-3\vec{i} + 3\vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ -3 & 0 & 3 \end{vmatrix} = 3\vec{i} - 6\vec{j} + 3\vec{k}$$

So a direction vector of L' is $\vec{i} - 2\vec{j} + \vec{k}$.

Equation of L':

$$x + 1 = \frac{y-3}{-2} = z + 2.$$

Example 10.6

Let $L : \dfrac{x-1}{2} = \dfrac{y-2}{-1} = \dfrac{z}{2}$ be a straight line and $\pi : x + y + z = 0$ be a plane in $\mathbb{R}^3$.

(c) Find the projection of L on π.

(c) Alternatively, a direction of vector of L' is given by

$$\begin{aligned}\vec{a} - \text{proj}_{\vec{n}}(\vec{a}) &= \vec{a} - \frac{\vec{a}\cdot\vec{n}}{|\vec{n}|^2}\vec{n} \\ &= 2\vec{i} - \vec{j} + 2\vec{k} - \frac{3}{3}(\vec{i} + \vec{j} + \vec{k}) \\ &= \vec{i} - 2\vec{j} + \vec{k}.\end{aligned}$$

Equation of L':

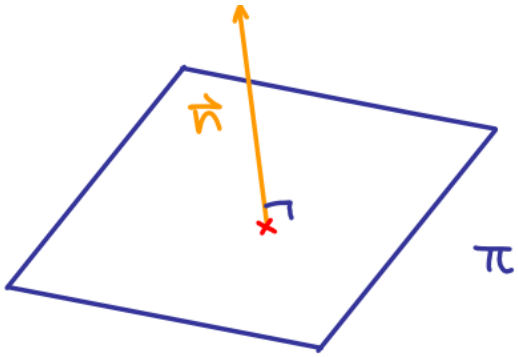
$$x + 1 = \frac{y-3}{-2} = z + 2.$$

Planes in $\mathbb{R}^3$

Follow the idea of the discussion of planes in $\mathbb{R}^3$, the equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n + b = 0$$

in $\mathbb{R}^n$ gives a "plane" in $\mathbb{R}^n$, which is said to be an affine hyperplane π. The vector $\vec{n} = (a_1, a_2, \ldots, a_n)$ is a normal of the affine hyperplane π.



π: $(n-1)$-dim affine hyperplane in $\mathbb{R}^n$, n-dim space.

- 1-dim affine hyperplane in $\mathbb{R}^2$ is just a usual straight line in $\mathbb{R}^2$
- 2-dim affine hyperplane in $\mathbb{R}^3$ is just a usual plane in $\mathbb{R}^3$

Curves

Idea: Suppose that a point is moving on xy-plane. $\gamma(t) = (x(t), y(t))$ is a function which tells us the x and y coordinate of the moving point at time t. Then the locus of the moving point is a curve in $\mathbb{R}^2$.

Definition 10.1

A **(parametric) curve** in $\mathbb{R}^n$ is a continuous function $\gamma : I \to \mathbb{R}^n$, where I is an interval. If we write

$$\gamma(t) = (x_1(t), x_2(t), \ldots, x_n(t)),\ t \in I,$$

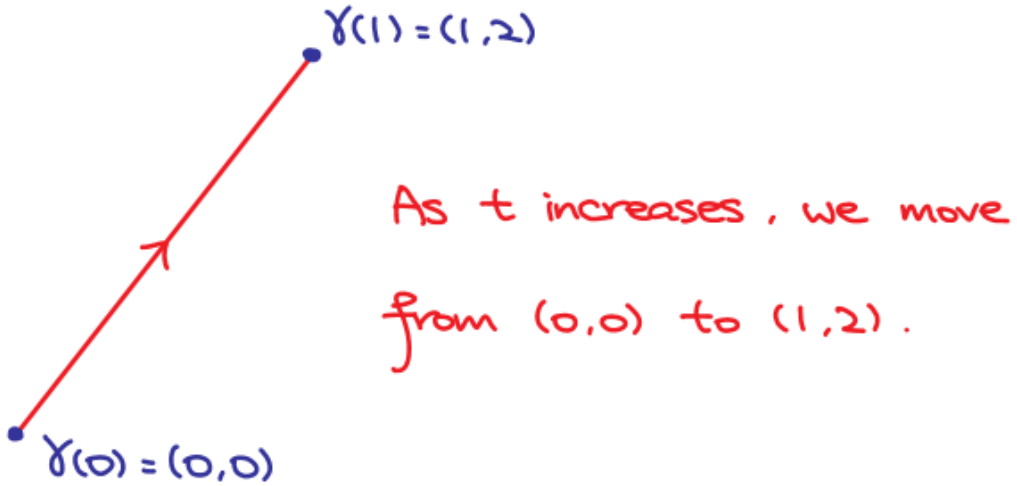
that means each component $x_i(t)$ is a continuous function.

Remark:
$\gamma(t)$ is a vector in $\mathbb{R}^n$, so sometimes γ is called a vector function and some may write $\vec{\gamma}(t)$.

Example 10.7

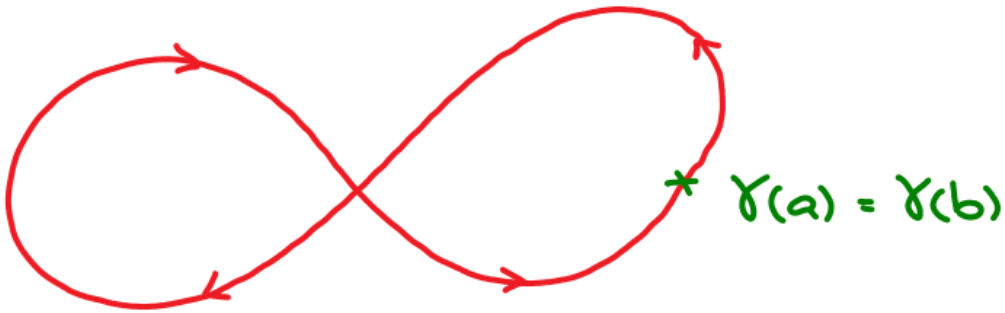
Let $\gamma : [0,1] \to \mathbb{R}^2$ be defined by $\gamma(t) = (x(t), y(t)) = (t, 2t)$.
Starting point: $\gamma(0) = (0,0)$
Endpoint: $\gamma(1) = (1,2)$

$$\begin{cases} x = t \\ y = t \end{cases} \xrightarrow{\text{eliminate } t} y = 2x$$



Definition 10.2

A curve $\gamma : [a, b] \to \mathbb{R}^n$ is said to be **closed** if $\gamma(a) = \gamma(b)$.

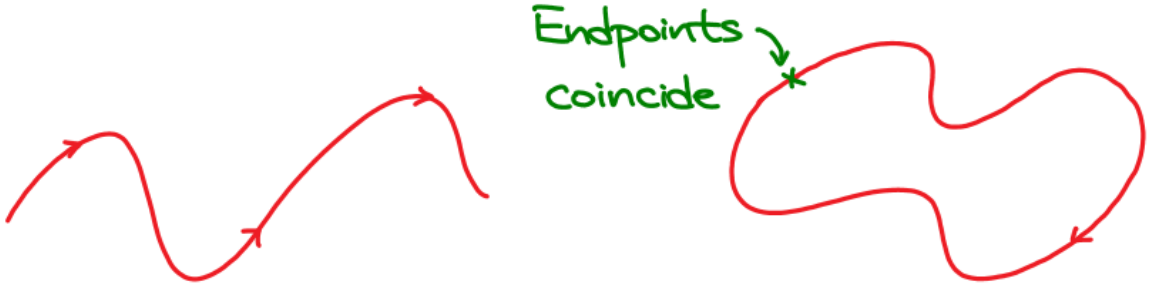


Definition 10.3

A curve $\gamma : [a, b] \to \mathbb{R}^n$ is said to be **simple** if it has no self intersection except possibly at its endpoints, i.e.

$$\gamma(t_1) \neq \gamma(t_2) \text{ for any } a \leq t_1 < t_2 \leq b, \text{ except possibly at } t_1 = a, t_2 = b.$$

NOT simple



Simple

Example 10.8

Let $\gamma : [0, 2\pi] \to \mathbb{R}^2$ be a curve defined by

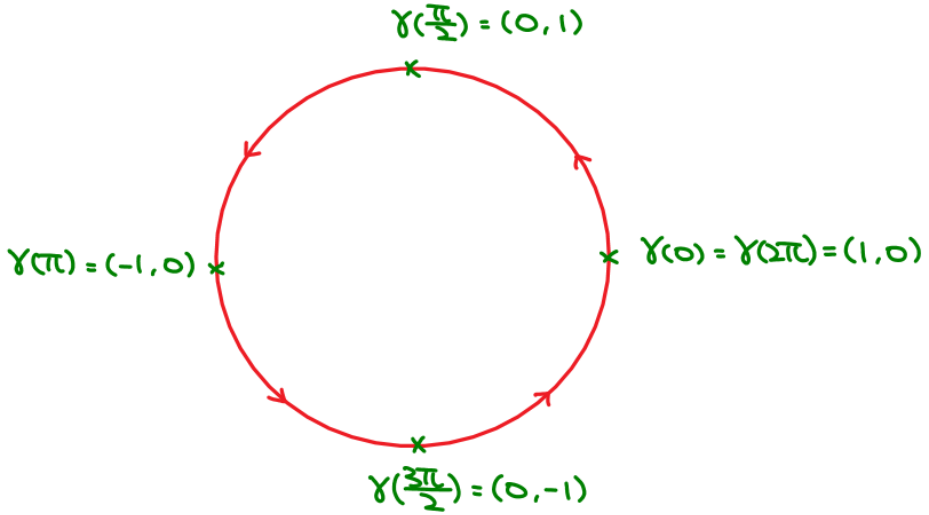
$$\gamma(t) = (R\cos t, R\sin t), \quad t \in [0, 2\pi]$$

where $R > 0$. Then $\gamma(0) = \gamma(2\pi) = (R, 0)$. Also,

$$\begin{cases} x = R\cos t & (1) \\ y = R\sin t & (2) \end{cases} \xrightarrow{\text{eliminate } t \text{ by } (1)^2 + (2)^2} \quad x^2 + y^2 = R^2$$

Therefore, γ is a circle centered at the origin with radius R.
As t increases, we move in anticlockwise direction.

$R = 1$:



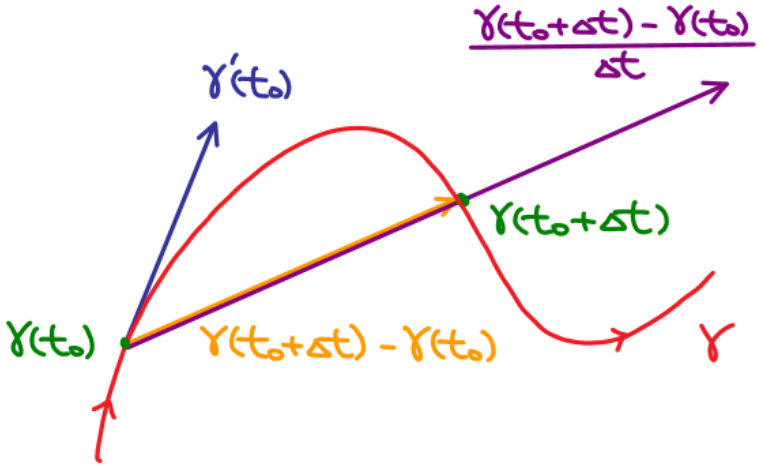
Definition 10.4

Let $\gamma : I \to \mathbb{R}^n$ be a curve. The derivative of γ at $t_0 \in I$ is defined as

$$\begin{aligned}
\gamma'(t_0) &= \lim_{\Delta t \to 0} \frac{\gamma(t_0 + \Delta t) - \gamma(t_0)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \left(\frac{x_1(t_0 + \Delta t) - x_1(t_0)}{\Delta t}, \ldots, \frac{x_n(t_0 + \Delta t) - x_n(t_0)}{\Delta t} \right) \\
&= \left(\lim_{\Delta t \to 0} \frac{x_1(t_0 + \Delta t) - x_1(t_0)}{\Delta t}, \ldots, \lim_{\Delta t \to 0} \frac{x_n(t_0 + \Delta t) - x_n(t_0)}{\Delta t} \right) \\
&= (x_1'(t_0), \ldots, x_n'(t_0)) \qquad \text{(if exists)}
\end{aligned}$$

If $\gamma'(t)$ exists for all $t \in I$, then γ is said to be a **differentiable curve**.

Remark: If t represents time and $\gamma(t)$ is the position of a moving object, then $\gamma'(t)$ and $|\gamma'(t)|$ are the velocity and speed of the object at time t respectively.



Example 10.8

Let $\gamma : [0, 2\pi] \to \mathbb{R}^2$ be a curve defined by $\gamma(t) = (R\cos t, R\sin t)$, where $R > 0, \omega$. Then

$$\gamma'(t) = (-R\sin t, R\cos t) \quad \text{and} \quad |\gamma'(t)| = \sqrt{(-R\sin t)^2 + (R\cos t)^2} = R.$$

Furthermore, let $\xi : [0, \frac{2\pi}{\omega}] \to \mathbb{R}^2$ be defined by $\xi(t) = (R\cos\omega t, R\sin\omega t)$, where $R > 0$.
Exercise: Check ξ also gives the same circle as γ but $|\xi'(t)| = R\omega$.
(Different parametrization may give the same curve.)

Exercise 10.2

Give a parametrization of a unit circle centered at the origin which goes in clockwise direction and starts and ends at $(0, 1)$.

Ans: One parametrization is $\gamma : [0, 2\pi] \to \mathbb{R}^2$,

$$\gamma(t) = (\cos(\frac{\pi}{2} - t), \sin(\frac{\pi}{2} - t)) = (\sin t, \cos t).$$

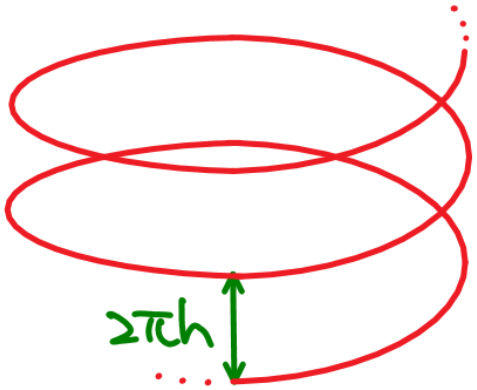
Exercise 10.3

Let $\gamma : \mathbb{R} \to \mathbb{R}^3$ be defined by

$$\gamma(t) = (x(t), y(t), z(t)) = (R\cos t, R\sin t, ht),$$

where $R, h > 0$. What is γ?

Ans: Helix:



Exercise 10.4

Give a parametrization of each of the following curves.

(1) Circle given by $(x-h)^2+(y-k)^2=R^2$, where $h,k\in\mathbb{R}$, $R>0$.

(2) Line segment joining two points A and B in $\mathbb{R}^n$ with $\overrightarrow{OA}=\vec{a}$ and $\overrightarrow{OB}=\vec{b}$.

[(3) Curve given by graph of a continuous function $f:[a,b]\to\mathbb{R}$.

Ans:

(1) $\gamma(t)=(h+R\cos t, k+R\sin t)$, $t\in[0,2\pi]$

(2) $\gamma(t)=\vec{a}+t(\vec{b}-\vec{a})=(1-t)\vec{a}+t\vec{b}$, $t\in[0,1]$

(3) $\gamma(t)=(t,f(t))$, $t\in[a,b]$